

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$y = 2x + 1$

$y = ax - 8$

In the system of equations above, a is a constant. If the system of equations has no solution, what is the value of a ?

Answer

A. $-\frac{1}{2}$

B. 0

C. 1

D. 2

Correct Answer: D

Rationale

Choice D is correct. A system of two linear equations has no solution when the graphs of the equations have the same slope and different y-coordinates of the y-intercepts. Each of the given equations is written in the slope-intercept form of a linear equation, $y = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept of the graph of the equation. For these two linear equations, the y-coordinates of the y-intercepts are different: **1** and **-8**. Thus, if the system of equations has no solution, the slopes of the two linear equations must be the same.

The slope of the first linear equation is 2. Therefore, for the system of equations to have no solution, the value of a must be 2.

Choices A, B, and C are incorrect and may result from conceptual and computational errors.